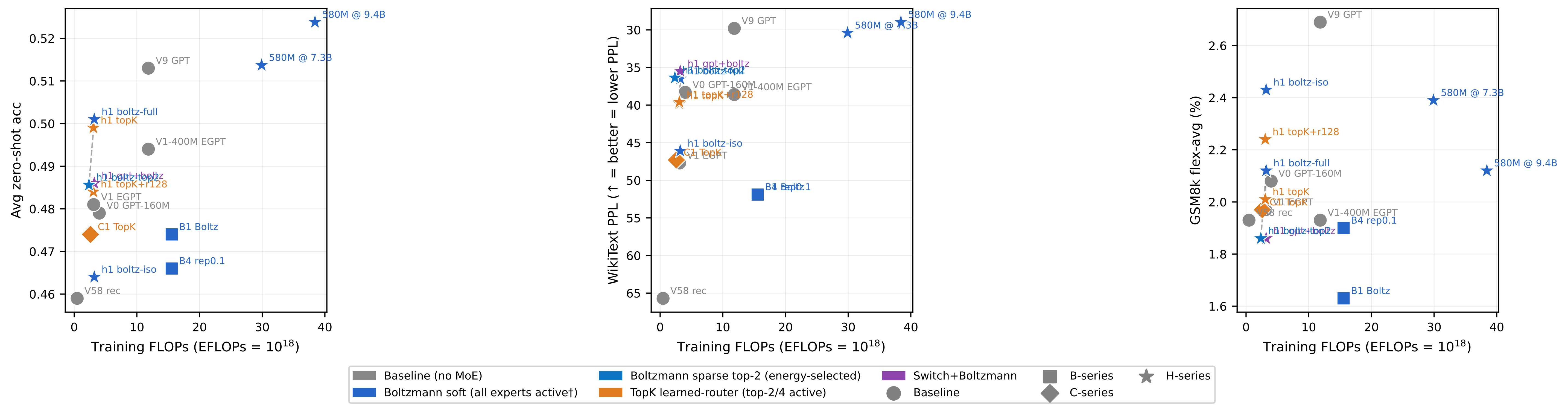
MoE variants vs baselines (7.86B tokens)
x-axis: training FLOPs ≈ 6 · active_params · tokens



FLOPs = 6 · active_params · tokens. Active params encode the sparsity assumption (Boltzmann soft = all experts active; sparse top-2 ≈ 75\% of soft).
Boltzmann routing itself is FLOPs-free vs the MoE forward — energy $E_i = x \cdot \text{term1}_i$ reuses term1 already needed for output.